KAREEM FAKHRY GAHEED ABD-ALLAH

Senior Data Analyst | Revenue, Pricing and Operations Analytics | SQL · Python · Power BI

Cairo, Egypt · +20 111 333 3744 · kareemfa5ry@gmail.com · linkedin.com/in/kareem-fakhry

PROFILE

Data analyst with 15 years in airline revenue, pricing and operations. Builds the full path from raw system files to a business decision: SQL and Python data pipelines, Power BI dashboards, forecasts, customer segmentation and statistical tests. Finds money that is being lost, such as uncollected penalties, unbilled extra baggage and partners breaking contract rules. Also trains commercial teams to use AI in their daily work.

SKILLS

SQL and databases: PostgreSQL, MySQL, DBeaver, joins and window functions, materialized views, indexing and query tuning, triggers

BI and reporting: Power BI, Power Query, Excel Data Model, star schema models with one-to-many relationships, Excel and VBA, automated daily reports

Statistics and analysis: Descriptive statistics (mean, median, spread), box plots and outlier checks, significance tests on samples, demand forecasting, customer segmentation (classification), churn analysis

Programming: Python (pandas, psycopg), web scraping, scheduled jobs, web dashboards (React, GraphQL, Redis)

AI: AI-assisted development (Cursor, VS Code, Windsurf), AI agents and automated workflows, AI training for business teams

Source systems: Amadeus (inventory, revenue accounting, billing files), Hitit, Accelya, SAP, Oracle

EXPERIENCE

Data and AI Consultant (freelance) | Nesma Airlines — Cairo · Feb 2026 – Present

- Kept on as a consultant after the employee role ended. Built a log-processing pipeline, a daily forecasting pipeline and an audit application between February and August 2026 (see Projects).
- Since August 2026, running a paid AI programme for the commercial department: a 5-day, 15-hour bilingual Arabic and English foundation course, then advanced sessions on AI agents and automated workflows. Each participant builds and presents a working workflow from their own daily tasks.

Revenue Manager | Nesma Airlines — Cairo · Oct 2022 – Jan 2026

Joined after new owners bought the airline, with reporting systems switched off. Built the revenue and data team from one person to five.

- Rebuilt commercial reporting by hand while systems were offline, confirming bookings and prices by phone with six sales agents, until systems returned three months later.
- Cleared a USD 1.5 million customer refund backlog before sales reopened. Settled about 60% without cash, through future incentives and discounted peak-season seat blocks given to selected top agents.
- Turned one manual sheet into an automated daily report: capacity, passengers, utilisation, average fare and revenue, actual against plan, by route, sales channel, currency and booking window.
- Built a 136-field transaction table, one row per ticket action, combining four sales sources, with every money item kept in both original and reporting currency.
- Evaluated a commercial revenue management system (PROS), found it did not fit the Egyptian market, and built an in-house forecast instead.
- January 2023 to January 2024: seats doubled while revenue tripled from USD 1.6 million to USD 5.0 million, with average fare up 37% and seat utilisation up from 71% to 81%.

Head of Revenue and Network Planning | Nile Air — Cairo · Jan 2017 – Jun 2021

Started as acting Revenue Manager and was confirmed after three months. Led four route controllers and yield executives.

- Repriced the full route network after the Egyptian currency devaluation.
- Brought fare filing in-house by securing a two-week ATPCO training programme, removing a paid outside dependency. ATPCO is the industry database where airlines file their fares.
- Took the Amadeus revenue accounting system through go-live with Finance and the vendor, fixing live data errors until reported figures could be trusted.
- Wrote the audit rules and procedures for a three-year outsourced revenue audit (Sutherland) covering sales, reissues, refunds and agent booking behaviour.
- Built the network and revenue data warehouse in SQL and Excel, and the five-year business plan prepared for the planned stock market listing.
- Decided with Planning which flights to cancel and which to operate through the COVID-19 closures and reopening.

Air Arabia | Egypt and UAE Dec 2010 – Jan 2016

Yield Management Executive (UAE, 2014–2016) · Revenue Management Support Executive (UAE, 2014) · Sales Support Executive (Egypt, 2012–2014) · Call Centre Representative (Egypt, 2010–2012)

- Priced routes, controlled seat inventory and overbooking, and analysed no-shows and group demand. Handled the Egypt base across fares, sales and promotions.
- Supported the market study for the Jordan base launch, and handled all contracted Umrah group bookings across three airports for a full season.

PROJECTS

Daily revenue data pipeline and forecasting dashboard Nesma Airlines, 2026

- Loads daily ticket and flight exports into PostgreSQL and applies only the rows that changed, found by comparing checksums. Runs in 17 stages that restart from the point of failure. A normal daily run takes about 9 minutes.
- Forecasts every flight departing in the next 14 days by comparing its bookings so far with similar past flights, then scores each flight and flags the ones that need a pricing or inventory action.
- Moved the heavy processing from Python into the database after the first version needed 35 GB of memory and stalled.

Stack: *PostgreSQL, SQL, materialized views, Python (psycopg), React, Apollo Client, GraphQL, Redis*

Operational log parser and revenue checks Nesma Airlines, 2026

- Parsed 68 million raw airline operational log records covering 17 months, and linked 11 million ticket events across separate systems.
- Matched 99% of 939,000 checked bags to the ticket that owned them, cutting unmatched records from 66% to under 1%. Found 40,981 paid extra-baggage records with no matching ticket, for revenue recovery.
- Confirmed 2,708 no-shows out of 3,779 cases checked, and set aside cases with conflicting boarding evidence so no passenger was wrongly charged.
- Made a key query about 660 times faster by adding a missing index. Protected results with locked tables, row fingerprints (SHA-256 hashing) and 193 automated tests.

Stack: *PostgreSQL, SQL, database triggers, Python, SHA-256 hashing, automated tests*

Customer and route analytics in Power BI Nesma Airlines

- Built dashboards for demand forecasting, customer segmentation by classification, personalised offers and customer churn.
- Used averages, medians and spread measures, box plots to find outliers, and significance tests on sampled routes to check whether differences between routes were real before acting on them.

Stack: *Power BI, Power Query, star schema data models (one-to-many relationships), SQL, Excel*

Revenue audit engine Nesma Airlines

- Checked every ticket against filed fare rules and contracted fees. Flagged refunds without the required penalty, refunds on non-refundable fares, late cancellations of issued tickets, and churning (an agent booking and cancelling the same seats again and again).
- Defined 38 standard breach categories and ran a manual check in parallel before relying on the automation.
- One quarter: 1,305 breaches across 170 sales partners, worth about USD 106,000. Finance accepted about 95% of the cases submitted.

Stack: *Excel, VBA, Power Query, Python (pandas), Amadeus billing files*

Competitor price tracker Nesma Airlines, 2023–2024

- Started as a script that captured fare calendar screenshots for the daily report. Grew into a scheduled scraper on company servers collecting one-way and 14-day return prices 90 days ahead for up to eight competing airlines per route.
- Cleaned data was emailed automatically every day to the commercial team, the revenue team and the chairman, and fed the pricing dashboards.

Stack: *Python, web scraping, Windows scheduled tasks, Excel, Power Query*

EDUCATION AND TRAINING

- BSc Accounting, Alexandria University, Faculty of Commerce (English section), 2010
- ATPCO fare filing and coding, two-week instructor-led programme, Nile Air head office
- Business Process Outsourcing Skills Programme, Information Technology Institute (ITI), nine months

LANGUAGES

Arabic (native) · English (full professional) · French (basic)